Annotation Bias Framework

This note formalizes how annotator-specific bias enters supervised training and how it affects reference-based evaluation. The central point is that a model trained on one annotator's labels targets that annotator's conditional mean, not the latent target. The two tasks below separate two related questions: whether self-reference makes the trained model appear artificially strong, and whether an observable proxy-reference comparison has the same sign as the comparison to the latent target.

1 Basic Setup

Let $X \sim P_X$ denote the data distribution. For an annotator identity ξ_r, the observed label from annotator r is modeled as

$$Y_r(x) = \mu_r(x) + \epsilon_r, \qquad \mu_r(x) = \mu^*(x) + b(x, \xi_r),$$

where $\mu^*(x)$ is the latent target, $b(x, \xi_r)$ is the systematic bias of annotator r on item x, and ϵ_r is idiosyncratic noise. We use the shorthand

$$b_r(x) := b(x, \xi_r).$$

Throughout, expectations $\mathbb{E}$ are taken over the data distribution without writing explicitly as $\mathbb{E}_X$.

The idiosyncratic noises are centered conditional on the item and annotator identity:

$$\mathbb{E}[\epsilon_r \mid X, \xi_r] = 0.$$

The annotator identities are independent and identically distributed draws from the annotator population. Conditional on the item and annotator identities, the observed labels from distinct annotators have independent idiosyncratic randomness:

$$Y_r \perp\!\!\!\perp Y_s \mid X, \xi_r, \xi_s, \qquad r \neq s.$$

For randomly drawn annotator identities, the bias field is centered itemwise:

$$\mathbb{E}_\xi[b(x, \xi)] = 0, \qquad \mathbb{E}_\xi[b(x, \xi)^2] = \tau_x^2 < \infty.$$

Thus τ_x^2 measures item-specific annotator variability.

2 ERM Link

Suppose the model is trained on annotator A's labels,

$$\{(x_i^{\mathrm{tr}}, y_{A,i}^{\mathrm{tr}})\}_{i=1}^n.$$

The empirical risk minimizer is

$$\hat{\mu}_A \in \arg\min_{g \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^{n} \{g(x_i^{\mathrm{tr}}) - y_{A,i}^{\mathrm{tr}}\}^2.$$

At the population level, the squared-loss risk under annotator A decomposes as

$$L_A(g) = \mathbb{E}\{g(X) - \mu_A(X)\}^2 + \mathbb{E}[\epsilon_A^2].$$

Therefore the population target of MSE training is μ_A, not μ^*. Writing

$$h_A(x) := \hat{\mu}_A(x) - \mu_A(x),$$

the trained model $\hat{\mu}_A(x)$ admits the decomposition

$$\hat{\mu}_A(x) = \mu^*(x) + b_A(x) + h_A(x). \tag{1}$$

The term h_A collects approximation and estimation error. If $\mu_A^{\mathcal{F}} \in \arg\min_{g \in \mathcal{F}} L_A(g)$ denotes the best-in-class population approximation, then

$$h_A = (\hat{\mu}_A - \mu_A^{\mathcal{F}}) + (\mu_A^{\mathcal{F}} - \mu_A).$$

Under the boundedness condition $Y_A, g \in [0,1]$, a standard Rademacher bound gives, with probability at least $1-\delta$,

$$\|h_A\|_2^2 \le \inf_{g \in \mathcal{F}} \|g - \mu_A\|_2^2 + 8\mathfrak{R}_n(\mathcal{F}) + 2\sqrt{\frac{\log(2/\delta)}{2n}}. \tag{2}$$

Consequently, unless both approximation and estimation errors vanish, the trained model inherits annotator A's systematic bias and may also carry non-negligible training error.

3 Reference-Based Comparison

For any reference source R, define the population comparison

$$\Delta_R := \mathbb{E}\Big[\{\hat{\mu}_A(X) - Y_R(X)\}^2 - \{Y_B(X) - Y_R(X)\}^2\Big].$$

The sign convention is

$$\Delta_R > 0 \quad \Longleftrightarrow \quad \hat{\mu}_A \text{ is worse than } B \text{ under reference } R. \tag{3}$$

On a held-out evaluation set of size m, the paired empirical statistic is

$$\hat{\Delta}_R := \frac{1}{m} \sum_{j=1}^{m} \Big[\{\hat{\mu}_A(x_j^{\mathrm{ev}}) - y_{R,j}^{\mathrm{ev}}\}^2 - \{y_{B,j}^{\mathrm{ev}} - y_{R,j}^{\mathrm{ev}}\}^2\Big].$$

All tests below use held-out observations. The data used to train $\hat{\mu}_A$ are separated from the data used for evaluation, so the paired statistics can be analyzed conditionally on the trained model.

4 Task 1: Self-Reference Exaggeration

4.1 Population Formulation

Task 1 asks whether using annotator A itself as the reference exaggerates the apparent performance of $\hat{\mu}_A$ relative to annotator B. Let C be another annotator and define

$$S_{A,C} := \Delta_C - \Delta_A.$$

Expanding the two reference comparisons gives

$$\begin{aligned} S_{A,C} &= \mathbb{E}\Big[\{\hat{\mu}_A - Y_C\}^2 - \{Y_B - Y_C\}^2 - \{\hat{\mu}_A - Y_A\}^2 + \{Y_B - Y_A\}^2\Big] \\ &= 2\mathbb{E}\big[(Y_A - Y_C)(\hat{\mu}_A - Y_B)\big]. \end{aligned}$$

Thus

$$S_{A,C} > 0 \quad \Longleftrightarrow \quad \Delta_A < \Delta_C. \tag{4}$$

A positive value means that $\hat{\mu}_A$ looks better when evaluated under self-reference A than when evaluated under the external reference C.

Substituting the model decomposition and using the centered idiosyncratic-noise conditions yields

$$\begin{aligned} \frac{1}{2}S_{A,C} &= \mathbb{E}\big[(b_A - b_C)(b_A - b_B + h_A)\big] \\ &= \mathbb{E}[b_A^2] - \mathbb{E}[b_A b_B] - \mathbb{E}\big[b_C(b_A - b_B)\big] + \mathbb{E}\big[(b_A - b_C)h_A\big]. \end{aligned}$$

The final term is the ERM-error remainder. In the ideal population ERM limit $h_A = 0$, if we consider A, B, C to be randomly drawn from the annotator population, then the itemwise centering and second-moment assumptions imply

$$\mathbb{E}_{A,B,C}[S_{A,C}] = 2\mathbb{E}_X[\tau_X^2] > 0.$$

If we keep the ERM error term h_A and assume that h_A is independent of the randomly drawn reference bias b_C, then the random-annotator expectation is perturbed only through the alignment between b_A and h_A. Combining this observation with the ERM bound in (2) gives

$$\big|\mathbb{E}_{A,B,C}S_{A,C} - 2\mathbb{E}_X[\tau_X^2]\big| \le 2\mathbb{E}_A\left[\inf_{g\in\mathcal{F}} \|g - \mu_A\|_2^2 + 8\mathfrak{R}_n(\mathcal{F}) + 2\sqrt{\frac{\log(2/\delta)}{2n}}\right]^{1/2}.$$

Remark 1. *The above analysis implies that, when treating A, B, C as random annotators drawn form the model, the exaggeration quantity $S_{A,C}$ is expected to be positive. However, for given annotators A, B, C, we still need a statistical test to verify (4).*

Remark 2. *Task I only tests whether exaggeration exists, but does not test whether the exaggeration causes an opposite conclusion as in task II.*

4.2 Statistic and Test

On a held-out evaluation set of size m, define

$$\hat{S}_{A,C} := \frac{2}{m}\sum_{j=1}^{m}(y_{A,j}^{\text{ev}} - y_{C,j}^{\text{ev}})\{\hat{\mu}_A(x_j^{\text{ev}}) - y_{B,j}^{\text{ev}}\}.$$

The one-sided hypotheses are

$$H_0 : S_{A,C} \le 0, \qquad H_1 : S_{A,C} > 0.$$

Let $\widehat{\mathrm{se}}(\hat{S}_{A,C})$ be the sample standard deviation of the paired summands divided by $\sqrt{m}$, and define

$$T_{A,C} := \frac{\hat{S}_{A,C}}{\widehat{\mathrm{se}}(\hat{S}_{A,C})}.$$

Reject H_0 at level α if

$$T_{A,C} > t_{m-1,1-\alpha}. \tag{5}$$

Theorem 1 (Asymptotic validity of Task 1)**.** *Condition on the trained model $\hat{\mu}_A$. Assume the held-out evaluation observations are independent and the paired difference $2(Y_A^{\mathrm{ev}} - Y_C^{\mathrm{ev}})(\hat{\mu}_A(X^{\mathrm{ev}}) - Y_B^{\mathrm{ev}})$ has finite nonzero variance. Under any fixed data-generating distribution satisfying $H_0 : S_{A,C} \le 0$, the one-sided paired test in* (5) *satisfies*

$$\limsup_{m\to\infty} \mathbb{P}(\textit{reject } H_0) \le \alpha.$$

Proof. Condition on $\hat{\mu}_A$. By the paired central limit theorem,

$$\frac{\hat{S}_{A,C} - S_{A,C}}{\mathrm{se}(\hat{S}_{A,C})} \Rightarrow N(0,1).$$

The usual sample-variance estimator is consistent under the same held-out sampling condition, so Slutsky's theorem gives

$$\frac{\hat{S}_{A,C} - S_{A,C}}{\widehat{\mathrm{se}}(\hat{S}_{A,C})} \Rightarrow N(0,1).$$

At the boundary $S_{A,C} = 0$, this is exactly $T_{A,C} \Rightarrow N(0,1)$, so the rejection probability converges to α. For any fixed $S_{A,C} < 0$, the numerator is shifted negatively by an order-one amount while the standard error is of order $m^{-1/2}$, so the rejection probability converges to zero. Therefore, under any fixed distribution satisfying H_0,

$$\limsup_{m\to\infty} \mathbb{P}(\text{reject } H_0) \le \alpha.$$

□

5 Task 2: Proxy-Reference Certification

5.1 Population Target and Sign Preservation

The ideal comparison against the latent target is

$$\Delta^* := \mathbb{E}\Big[\{\hat{\mu}_A(X) - \mu^*(X)\}^2 - \{Y_B(X) - \mu^*(X)\}^2\Big].$$

The observable comparison against reference R is Δ_R. Define the proxy distortion

$$g_R := \mathbb{E}\big[(Y_R - \mu^*)(\hat{\mu}_A - Y_B)\big].$$

Then we can verify that

$$\Delta_R - \Delta^* = -2g_R. \tag{6}$$

Thus the proxy comparison preserves the same nonzero sign as the ideal comparison if and only if

$$\Delta_R(\Delta_R + 2g_R) > 0.$$

Task 2 therefore asks whether the observable proxy signal is large enough to dominate the possible sign-reversing distortion induced by the reference source. Let R contain K reference annotators, disjoint from A and B. The reference label is defined to be the average

$$Y_R(x) = \frac{1}{K}\sum_{r\in R} Y_r(x), \qquad K = |R|. \tag{7}$$

Therefore the proxy distortion can be written as the average of the annotator-level contributions

$$g_R = \frac{1}{K}\sum_{r\in R} g_r, \qquad g_r := \mathbb{E}\big[(Y_r - \mu^*)(\hat{\mu}_A - Y_B)\big].$$

Under the conditional independence and conditional mean-zero noise assumptions in the Basic Setup, each contribution satisfies

$$g_r = \mathbb{E}_X\big[b_r(X)\{\hat{\mu}_A(X) - \mu_B(X)\}\big].$$

Thus Task 2 controls the full proxy distortion directly, rather than reducing it to a separate global reference offset and a separate mean gap between $\hat{\mu}_A$ and Y_B.

5.2 Annotator-Level Proxy-Distortion Model

Conditioning on the trained model $\hat{\mu}_A$ and on annotator B, the randomness in g_r is induced by the reference annotator identity. We assume the annotator-level proxy distortions satisfy

$$g_r \overset{\text{i.i.d.}}{\sim} N(0, \tau_g^2). \tag{8}$$

The zero mean is consistent with itemwise centering, since

$$\mathbb{E}_r[g_r] = \mathbb{E}_X\Big[\mathbb{E}_r\{b_r(X)\}\{\hat{\mu}_A(X) - \mu_B(X)\}\Big] = 0.$$

The reference-set distortion therefore satisfies

$$g_R = \frac{1}{K}\sum_{r\in R} g_r \sim N\left(0, \frac{\tau_g^2}{K}\right).$$

Remark 3. *The Gaussian random-effect assumption in* (8) *is stronger than the Basic Setup and is specific to Task 2. One finite-sample motivation is the following. Conditional on $\hat{\mu}_A$ and annotator B, define $d_B(x) = \hat{\mu}_A(x) - \mu_B(x)$. If, for any finite collection $x_1, \ldots, x_n$, the vector*

$$(b_r(x_1), \ldots, b_r(x_n))$$

is jointly Gaussian over the annotator population, then

$$\frac{1}{n}\sum_{i=1}^{n} b_r(x_i) d_B(x_i)$$

is a linear combination of a jointly Gaussian vector and is therefore Gaussian. The population-level normal model for g_r is the corresponding random-effect idealization.

5.3 Calibration of τ_g on D_2

Let D_2 denote the observations used for auxiliary calibration. The notation D_1 and D_2 only separates the main-statistic and calibration roles; it imposes no requirement between them, and the two sets may even coincide. By the last sub-section, it is enough to calibrate τ_g in order to control g_R. Suppose $|R| > 1$. Choose a baseline reference annotator indexed by 1. For $r = 2, \ldots, K$, define the observable weighted difference

$$w_i^r := (y_{ir} - y_{i1})\{\hat{\mu}_A(x_i) - y_{B,i}\}.$$

The latent target cancels in the difference:

$$\mathbb{E}[w_i^r] = \mathbb{E}[(Y_r - Y_1)(\hat{\mu}_A - Y_B)] = g_r - g_1.$$

Define

$$\bar{w}_r := \frac{1}{|D_2|} \sum_{i \in D_2} w_i^r, \qquad \bar{\boldsymbol{w}} := (\bar{w}_2, \ldots, \bar{w}_K)^\top \in \mathbb{R}^{K-1}.$$

Under (8), the random-effect part has

$$\mathrm{Var}(g_r - g_1) = 2\tau_g^2, \qquad \mathrm{Cov}(g_r - g_1, g_s - g_1) = \tau_g^2, \quad r \neq s.$$

Let $\boldsymbol{\Omega}$ denote the covariance matrix of the weighted difference vector $(w_i^2, \ldots, w_i^K)^\top$, and let $\widehat{\boldsymbol{\Omega}}$ be its sample covariance on D_2. For a candidate value t of τ_g, the model-implied covariance matrix $\boldsymbol{V}_g(t)$ of $\bar{\boldsymbol{w}}$ has entries

$$\boldsymbol{V}_g(t)_{rr} = 2t^2 + \frac{\boldsymbol{\Omega}_{rr}}{|D_2|}, \qquad \boldsymbol{V}_g(t)_{rs} = t^2 + \frac{\boldsymbol{\Omega}_{rs}}{|D_2|}, \quad r \neq s.$$

In practice, use the plug-in matrix $\widehat{\boldsymbol{V}}_g(t)$ obtained by replacing $\boldsymbol{\Omega}$ with $\widehat{\boldsymbol{\Omega}}$:

$$\widehat{\boldsymbol{V}}_g(t)_{rr} = 2t^2 + \frac{\widehat{\boldsymbol{\Omega}}_{rr}}{|D_2|}, \qquad \widehat{\boldsymbol{V}}_g(t)_{rs} = t^2 + \frac{\widehat{\boldsymbol{\Omega}}_{rs}}{|D_2|}, \quad r \neq s,$$

because the pairwise differences share the baseline distortion g_1. The asymptotic chi-square calibration statistic is

$$\hat{Q}_g(t) := \bar{\boldsymbol{w}}^\top \widehat{\boldsymbol{V}}_g(t)^{-1} \bar{\boldsymbol{w}}.$$

Throughout this calibration step, K is treated as fixed as $|D_2| \to \infty$. Let $q_{\gamma,K-1}$ be the lower γ-quantile of χ^2_{K-1}. Since $\hat{Q}_g(t)$ is non-increasing in t, define the upper endpoint by lower-tail inversion:

$$U_{\tau_g,\gamma}(|D_2|) := \sup\{t \geq 0 : \hat{Q}_g(t) \geq q_{\gamma,K-1}\}. \tag{9}$$

The resulting calibration property is

$$\mathbb{P}\big(\tau_g \leq U_{\tau_g,\gamma}(|D_2|)\big) \geq 1 - \gamma + o(1), \qquad |D_2| \to \infty.$$

Combining this coverage property with the normal distribution of g_R gives

$$\mathbb{P}\left(|g_R| \leq z_{1-\alpha_g/2} \frac{U_{\tau_g,\gamma}(|D_2|)}{\sqrt{K}}\right) \geq 1 - \alpha_g - \gamma + o(1). \tag{10}$$

This fixed-K calibration is the regime used here; allowing K to grow would require additional high-dimensional covariance consistency conditions.

Proof. At $t = \tau_g$, the model-implied covariance of $\bar{\boldsymbol{w}}$ is $\boldsymbol{V}_g(\tau_g)$. Under the fixed-K multivariate CLT for the calibration averages,

$$\boldsymbol{V}_g(\tau_g)^{-1/2}\bar{\boldsymbol{w}} \Rightarrow N(0, \boldsymbol{I}_{K-1}),$$

and therefore the quadratic form with the true covariance satisfies

$$\bar{\boldsymbol{w}}^\top \boldsymbol{V}_g(\tau_g)^{-1}\bar{\boldsymbol{w}} \Rightarrow \chi^2_{K-1}.$$

If $\widehat{\boldsymbol{\Omega}} \xrightarrow{p} \boldsymbol{\Omega}$, then $\widehat{\boldsymbol{V}}_g(\tau_g) \xrightarrow{p} \boldsymbol{V}_g(\tau_g)$. Since K is fixed, Slutsky's theorem gives

$$\hat{Q}_g(\tau_g) = \bar{\boldsymbol{w}}^\top \widehat{\boldsymbol{V}}_g(\tau_g)^{-1}\bar{\boldsymbol{w}} \Rightarrow \chi^2_{K-1}.$$

Because $\hat{Q}_g(t)$ is non-increasing in t,

$$\{\tau_g \le U_{\tau_g,\gamma}(|D_2|)\} = \{\hat{Q}_g(\tau_g) \ge q_{\gamma,K-1}\}.$$

Hence

$$\mathbb{P}\big(\tau_g \le U_{\tau_g,\gamma}(|D_2|)\big) = \mathbb{P}\big(\hat{Q}_g(\tau_g) \ge q_{\gamma,K-1}\big) = 1 - \gamma + o(1).$$

Finally, since $g_R \sim N(0, \tau_g^2/K)$,

$$\mathbb{P}\left(|g_R| \le z_{1-\alpha_g/2}\frac{\tau_g}{\sqrt{K}}\right) = 1 - \alpha_g.$$

On the intersection of this event and $\{\tau_g \le U_{\tau_g,\gamma}(|D_2|)\}$,

$$|g_R| \le z_{1-\alpha_g/2}\frac{\tau_g}{\sqrt{K}} \le z_{1-\alpha_g/2}\frac{U_{\tau_g,\gamma}(|D_2|)}{\sqrt{K}}.$$

A union bound therefore gives the displayed probability bound. □

5.4 Main Statistic on D_1

On the main evaluation set D_1, define

$$\hat{\Delta}_R := \frac{1}{|D_1|}\sum_{j\in D_1}\Big[\{\hat{\mu}_A(x_j) - y_{R,j}\}^2 - \{y_{B,j} - y_{R,j}\}^2\Big],$$

and assume the paired CLT and standard-error consistency

$$\frac{\hat{\Delta}_R - \Delta_R}{\widehat{\mathrm{se}}(\hat{\Delta}_R)} \Rightarrow N(0,1), \qquad \frac{\widehat{\mathrm{se}}(\hat{\Delta}_R)}{\mathrm{se}(\hat{\Delta}_R)} \xrightarrow{p} 1.$$

5.5 Certification Test

The certification null is that the proxy comparison and the ideal comparison do not have the same strictly nonzero sign:

$$H_0 : \Delta_R(\Delta_R + 2g_R) \le 0, \qquad H_1 : \Delta_R(\Delta_R + 2g_R) > 0.$$

Reject H_0 if

$$|\hat{\Delta}_R| - t_{|D_1|-1,1-\alpha_\Delta/2}\widehat{\mathrm{se}}(\hat{\Delta}_R) > 2z_{1-\alpha_g/2}\frac{U_{\tau_g,\gamma}(|D_2|)}{\sqrt{K}}. \tag{11}$$

The left-hand side is a lower confidence bound for the magnitude of the observable proxy signal. The right-hand side is an upper bound for the possible sign-reversing distortion under the annotator-level proxy-distortion model.

Theorem 2 (Asymptotic validity of Task 2)**.** *Assume the calibration coverage property with endpoint $U_{\tau_g,\gamma}(|D_2|)$. Assume also the paired CLT and standard-error consistency for $\hat{\Delta}_R$. Under any fixed data-generating distribution satisfying H_0, the certification rule satisfies*

$$\limsup_{|D_1|,|D_2|\to\infty} \mathbb{P}(\textit{reject } H_0) \le \alpha_\Delta + \alpha_g + \gamma.$$

Proof. Work under a fixed data-generating distribution satisfying H_0. All $o(1)$ terms in this proof are taken along the joint limit $|D_1|, |D_2| \to \infty$. By the paired CLT for $\hat{\Delta}_R$,

$$\mathbb{P}\left(|\hat{\Delta}_R - \Delta_R| \le t_{|D_1|-1,1-\alpha_\Delta/2}\widehat{\mathrm{se}}(\hat{\Delta}_R)\right) \ge 1 - \alpha_\Delta + o(1).$$

On this event, the reverse triangle inequality gives

$$|\hat{\Delta}_R| - t_{|D_1|-1,1-\alpha_\Delta/2}\widehat{\mathrm{se}}(\hat{\Delta}_R) \le |\Delta_R|.$$

If H_0 holds, then

$$\Delta_R(\Delta_R + 2g_R) \le 0.$$

Thus Δ_R and $\Delta_R + 2g_R$ have opposite signs or one of them is zero. Hence 0 lies between them, which implies

$$|\Delta_R| \le 2|g_R|.$$

Combining the preceding lower bound, the implication under H_0, and the rejection rule, rejection on the high-probability CLT event implies

$$|g_R| > z_{1-\alpha_g/2}\frac{U_{\tau_g,\gamma}(|D_2|)}{\sqrt{K}}.$$

On the calibration event $\{\tau_g \le U_{\tau_g,\gamma}(|D_2|)\}$, and using $g_R/(\tau_g/\sqrt{K}) \sim N(0,1)$, the probability of this event is at most $\alpha_g + o(1)$. Combining the CLT event, calibration coverage, and the normal tail bound by a union bound, without requiring independence between these events, gives

$$\mathbb{P}(\text{reject } H_0) \le \alpha_\Delta + \gamma + \alpha_g + o(1).$$

Taking the limsup proves the result. □

5.6 Procedure

The complete Task 2 certification procedure is as follows.

1. Train $\hat{\mu}_A$ on annotator A's training labels, using data disjoint from D_1 and D_2.
2. Draw a reference set R of size K from the annotator population, with $R \cap \{A, B\} = \varnothing$.
3. On D_2, compute the weighted differences $(y_{ir} - y_{i1})\{\hat{\mu}_A(x_i) - y_{B,i}\}$, form the calibration statistic $\hat{Q}_g(t)$, and obtain $U_{\tau_g,\gamma}(|D_2|)$ satisfying the stated coverage property.
4. On D_1, compute $\hat{\Delta}_R$ and $\widehat{\mathrm{se}}(\hat{\Delta}_R)$.
5. Certify the proxy sign if the rejection condition holds.

6 Effect of the Reference-Set Size K

The role of K is determined by the rejection rule in (11):

$$|\hat{\Delta}_R| - t_{|D_1|-1,1-\alpha_\Delta/2}\widehat{\mathrm{se}}(\hat{\Delta}_R) > 2z_{1-\alpha_g/2}\frac{U_{\tau_g,\gamma}(|D_2|)}{\sqrt{K}}.$$

For fixed $\alpha_\Delta, \alpha_g, \gamma$, the quantile $z_{1-\alpha_g/2}$ is constant. The calibration endpoint $U_{\tau_g,\gamma}(|D_2|)$ estimates the annotator-level proxy-distortion scale τ_g. Under the calibration regime in Section 5.3, if $\tau_g = O(1)$ and the weighted-difference covariance matrix remains of the same order, then

$$U_{\tau_g,\gamma}(|D_2|) = O_p(1).$$

Therefore the reference-bias correction in the rejection rule satisfies

$$2z_{1-\alpha_g/2}\frac{U_{\tau_g,\gamma}(|D_2|)}{\sqrt{K}} = O_p(K^{-1/2}). \tag{12}$$

This aligns with the decay rate of the population distortion g_R, so increasing K shrinks the unobserved sign-reversing distortion that the right-hand side of (11) must dominate.

Now fix D_1 and let $K \to \infty$. The averaged reference label approaches the ideal reference μ^*, so the left-hand side approaches the ideal-reference lower confidence signal. Define

$$\hat{\Delta}^* := \frac{1}{|D_1|}\sum_{j\in D_1}\left[\{\hat{\mu}_A(x_j) - \mu^*(x_j)\}^2 - \{y_{B,j} - \mu^*(x_j)\}^2\right].$$

Then

$$|\hat{\Delta}_R| - t_{|D_1|-1,1-\alpha_\Delta/2}\widehat{\mathrm{se}}(\hat{\Delta}_R) \longrightarrow |\hat{\Delta}^*| - t_{|D_1|-1,1-\alpha_\Delta/2}\widehat{\mathrm{se}}(\hat{\Delta}^*).$$

Thus, for fixed D_1, increasing K does not make the left-hand side diverge. It makes the reference-based lower confidence signal approach its ideal-reference counterpart, while the right-hand side decreases to zero at rate $K^{-1/2}$. If the limiting left-hand side is positive, then sufficiently large K makes certification possible. If it is zero or negative, increasing K alone cannot guarantee certification because the fixed D_1 signal is not strong enough relative to its sampling penalty.

Remark 4. *If $|D_1| \to \infty$ as well, then $\hat{\Delta}^* \to \Delta^*$ and $\widehat{\mathrm{se}}(\hat{\Delta}^*) \to 0$. Hence, when the ideal signal is strictly nonzero, the limiting left-hand side is eventually positive. This implies that, given sufficient observations, the reference set R would eventually satisfy the condition of rejecting H_0 as K constantly goes large.*

Remark 5. *The test in Task 2 makes sense only when the reference annotators are treated as random draws from a larger annotator population, with the annotator-level proxy distortions modeled as in* (8).

7 Summary of the Framework

The framework separates three effects. First, ERM trained on annotator A targets $\mu_A = \mu^* + b_A$ rather than μ^*, so $\hat{\mu}_A$ inherits systematic annotation bias. Second, Task 1 detects whether evaluating $\hat{\mu}_A$ under self-reference A exaggerates its apparent performance relative to an external annotator. The statistic $S_{A,C}$ is observable and admits an asymptotically valid paired test. Third, Task 2 certifies whether an observable reference comparison Δ_R has the same sign as the ideal comparison Δ^*. Under conditional independence across annotators given the item and annotator identities, an annotator-level Gaussian model for the weighted proxy distortions g_r, and valid calibration of τ_g, the certification rule controls false certification at level $\alpha_\Delta + \alpha_g + \gamma + o(1)$.

8 Limitation of reference set

The Task 2 formulation above is written for a reference set R with observed reference labels Y_r. In the current dataset, each item has only three observed annotators, so a large observed reference average cannot be constructed directly:

$$Y_R(x) = \frac{1}{K} \sum_{r \in R} Y_r(x).$$

For Task 2 only, replace each observed reference annotator by a trained reference model. For each $r \in R$, train $\hat{\mu}_r$ on annotator r's labels and use the same ERM decomposition as for annotator A:

$$\hat{\mu}_r(x) = \mu^*(x) + b_r(x) + h_r(x).$$

Define the trained reference average

$$\hat{\mu}_R(x) = \frac{1}{K} \sum_{r \in R} \hat{\mu}_r(x).$$

Under this dataset limitation, the Task 2 reference source should be changed by the replacement

$$Y_R \longrightarrow \hat{\mu}_R.$$

Thus the population comparison and proxy distortion become

$$\Delta_{\hat{R}} := \mathbb{E}\Big[\{\hat{\mu}_A(X) - \hat{\mu}_R(X)\}^2 - \{Y_B(X) - \hat{\mu}_R(X)\}^2\Big],$$

$$g_{\hat{R}} := \mathbb{E}\big[\{\hat{\mu}_R(X) - \mu^*(X)\}\{\hat{\mu}_A(X) - Y_B(X)\}\big].$$

The same algebraic identity remains:

$$\Delta_{\hat{R}} - \Delta^* = -2g_{\hat{R}}.$$

Equivalently,

$$g_{\hat{R}} = \frac{1}{K} \sum_{r \in R} g_{\hat{r}}, \qquad g_{\hat{r}} := \mathbb{E}\big[\{\hat{\mu}_r(X) - \mu^*(X)\}\{\hat{\mu}_A(X) - Y_B(X)\}\big].$$

Let

$$\bar{h}_R(x) := \frac{1}{K} \sum_{r \in R} h_r(x).$$

The trained-reference version requires the following two assumptions.

Assumption 1 (Centered trained-reference error)**.**

$$\mathbb{E}[\bar{h}_R(X)] = 0.$$

Assumption 2 (Orthogonality of trained-reference error)**.**

$$\bar{h}_R(X) \perp\!\!\!\perp \{\hat{\mu}_A(X) - Y_B(X)\}.$$

These assumptions make the trained-reference estimation error contribute zero to the Task 2 distortion:

$$\mathbb{E}\big[\bar{h}_R(X)\{\hat{\mu}_A(X) - Y_B(X)\}\big] = 0.$$

Therefore the trained-reference distortion has the same target as the observed-reference distortion after the replacement $Y_R \to \hat{\mu}_R$. The normal random-effect model is applied directly to the trained-reference contributions $g_{\hat{r}}$, using the same proxy-distortion scale τ_g:

$$g_{\hat{r}} \overset{\text{i.i.d.}}{\sim} N(0, \tau_g^2), \qquad g_{\hat{R}} \sim N\left(0, \frac{\tau_g^2}{K}\right).$$

No separate global-offset correction is introduced, and no additional correction term is added.

On D_1, the main statistic should be implemented as

$$\hat{\Delta}_{\hat{R}} := \frac{1}{|D_1|} \sum_{j \in D_1} \Big[\{\hat{\mu}_A(x_j) - \hat{\mu}_R(x_j)\}^2 - \{y_{B,j} - \hat{\mu}_R(x_j)\}^2\Big].$$

The standard error is computed from these paired summands. The rejection rule becomes

$$|\hat{\Delta}_{\hat{R}}| - t_{|D_1|-1,1-\alpha_\Delta/2}\widehat{\text{se}}(\hat{\Delta}_{\hat{R}}) > 2z_{1-\alpha_g/2}\frac{U_{\tau_g,\gamma}(|D_2|)}{\sqrt{K}}.$$

For calibration on D_2, replace the observed weighted differences with trained-reference weighted prediction differences:

$$w_i^r = \{\hat{\mu}_r(x_i) - \hat{\mu}_1(x_i)\}\{\hat{\mu}_A(x_i) - y_{B,i}\}, \qquad r = 2, \ldots, K.$$

Under Assumptions 1 and 2, these differences target the trained-reference distortion contrasts:

$$\mathbb{E}[w_i^r] = g_{\hat{r}} - g_{\hat{1}}.$$

The calibration step therefore estimates the same proxy-distortion scale τ_g used in Section 5.3, with the full sample covariance matrix of the trained-reference weighted difference vector.

Finally, the Task 2 null and alternative should only replace (Δ_R, g_R) with $(\Delta_{\hat{R}}, g_{\hat{R}})$:

$$H_0 : \Delta_{\hat{R}}(\Delta_{\hat{R}} + 2g_{\hat{R}}) \le 0, \qquad H_1 : \Delta_{\hat{R}}(\Delta_{\hat{R}} + 2g_{\hat{R}}) > 0.$$

The validity statement of Theorem 2 is unchanged:

$$\limsup_{|D_1|,|D_2|\to\infty} \mathbb{P}(\text{reject } H_0) \le \alpha_\Delta + \alpha_g + \gamma.$$

The proof follows the original Task 2 proof after applying the replacements $Y_R \to \hat{\mu}_R$, $\Delta_R \to \Delta_{\hat{R}}$, $g_R \to g_{\hat{R}}$, and $\hat{\Delta}_R \to \hat{\Delta}_{\hat{R}}$.